# Output-error sensitivity of a Clements mesh vs. effective insertion-loss spread (large range, exploratory; fixed $\mu_L = 1$ dB, varying $N$)

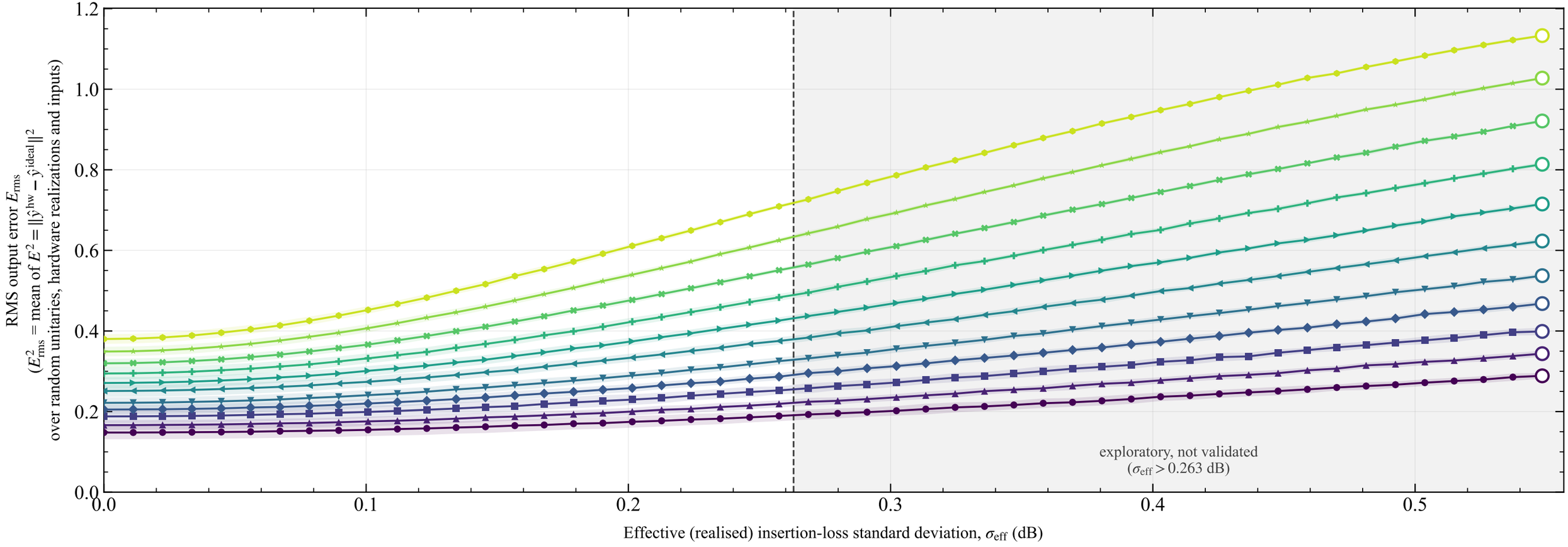
$N = 20, 28, 38, 53, 72, 100, 138, 190, 263, 362, 500$ | Haar-random unitary $U$ (single Clements mesh) | one forward pass $y = Ux$ | $\hat{y}$ = complex output field, normalised to unit L2 norm | $E = \|\hat{y}^{\mathrm{hw}} - \hat{y}^{\mathrm{ideal}}\|_2$

Inputs per unitary: 7 types x 10 random draws (dense complex, dense real, binary ±1, phase-only, sparse 10%, single port, Gaussian beam) | shown: mean over all 7 types (equal weight)

30 hardware realizations per unitary per point (each chip sees one input of every type; each input used on 3 chips) | 25 Haar unitaries | 50 points per curve in $\sigma_{\mathrm{eff}} \in [0, \sigma_{\mathrm{top}}(\mu_L)]$ dB | curves at fixed $\mu_L$ = 1 dB, one per network size

Hardware model: per-arm loss $L \sim \mathrm{TruncNorm}(\mu_L, \sigma_L^2; 0, 2\mu_L)$ dB, amplitude $10^{-L/20}$, independent for each arm of every MZI and chip; swept: realised std $\sigma_{\mathrm{eff}}$ of $L$ ($\sigma_L$ by inversion); single Clements mesh, ideal phase shifters, 50:50 couplers and output phase screen

Each point: $E_{\mathrm{rms}} = \sqrt{\langle E^2 \rangle}$ over unitaries, chips and input types; band: ±1 s.d. across Haar-random targets (s.e. in the CSV) | $\sigma_{\mathrm{eff}} = 0$: delta at $\mu_L$ (starts at $E_L(\mu_L, N)$) | $\sigma_{\mathrm{top}} = 0.95\,\mu_L/\sqrt{3}$, exploratory above 0.263 dB




Network size $N$ | $E_0 = E_{\mathrm{rms}}(\sigma_{\mathrm{eff}} = 0)$, $E_{\max} = E_{\mathrm{rms}}$ at $\sigma_{\mathrm{top}}$ (open circle = end of curve); $\sigma_{\mathrm{top}} = 0.95\,\mu_L/\sqrt{3}$ (0.263 dB cap lifted)

- $N = 20$ ($E_0 = 0.148$, $E_{\max} = 0.289$)
- $N = 28$ ($E_0 = 0.166$, $E_{\max} = 0.344$)
- $N = 38$ ($E_0 = 0.188$, $E_{\max} = 0.399$)
- $N = 53$ ($E_0 = 0.205$, $E_{\max} = 0.468$)
- $N = 72$ ($E_0 = 0.222$, $E_{\max} = 0.537$)
- $N = 100$ ($E_0 = 0.251$, $E_{\max} = 0.623$)
- $N = 138$ ($E_0 = 0.271$, $E_{\max} = 0.715$)
- $N = 190$ ($E_0 = 0.295$, $E_{\max} = 0.814$)
- $N = 263$ ($E_0 = 0.320$, $E_{\max} = 0.921$)
- $N = 362$ ($E_0 = 0.349$, $E_{\max} = 1.028$)
- $N = 500$ ($E_0 = 0.380$, $E_{\max} = 1.133$)